

**Comme Alice,
osez suivre le lapin
blanc et plongez
dans la JVM pour
comprendre son
fonctionnement**



Qui suis-je ?



Antoine DESSAIGNE



Remerciements

- Aleksey Shipilëv
 - OpenJDK @ AWS
- Nicolai Parlog
 - Java Developer Advocate @ Oracle



Qu'est-ce qui se passe ?

- **NullPointerException**
 - sur 1 benchmark sur 16
 - seulement 50% des exécutions
- **Aucun problème sur nos tests**
 - environ 11000 tests unitaires
 - environ 6000 tests d'intégration
- **Vu 15 jours après avoir migré de 17.0.8 à 21.0.1**

Que fait ce benchmark ?

- Il mesure la vitesse de rattrapage
- on simule 30 minutes de données
- on mesure le temps pour réabsorber ces 30 minutes (environ 1 minute 30)

Et le code dans tout ça ?

```
final ReadableInterval validTime = value.getValidTime();
```

```
...
```

```
if (validTime.getStartMillis() < ...) {
```

```
}
```

Et le code dans tout ça ?

```
final ReadableInterval validTime = value.getValidTime();
```

```
...
```

```
if (validTime.getStartMillis() < ...) {  
    // NullPointerException  
    // Cannot invoke "ReadableInterval.getStartMillis()"   
    // because "validTime" is null  
}
```

Et le code dans tout ça ?

```
final ReadableInterval validTime = value.getValidTime();
```

```
...
```

```
if (validTime.getStartMillis() < ...) {  
    // NullPointerException  
    // Cannot invoke "ReadableInterval.getStartMillis()"   
    // because "validTime" is null  
}
```

Et le code dans tout ça ?

```
final ReadableInterval validTime = value.getValidTime();  
if (validTime.getStartMillis() < ...) {  
  
}
```

...

```
if (validTime.getStartMillis() < ...) {  
    // NullPointerException  
    // Cannot invoke "ReadableInterval.getStartMillis()"   
    // because "validTime" is null  
}
```

Et le code dans tout ça ?

```
final ReadableInterval validTime = value.getValidTime();  
if (validTime.getStartMillis() < ...) {  
  
}
```

...

```
if (validTime.getStartMillis() < ...) {  
    // NullPointerException  
    // Cannot invoke "ReadableInterval.getStartMillis()"   
    // because "validTime" is null  
}
```

Et le code dans tout ça ?

```
final ReadableInterval validTime = value.getValidTime();  
if (validTime.getStartMillis() < ...) {  
    // No exception here  
}
```

...

```
if (validTime.getStartMillis() < ...) {  
    // NullPointerException  
    // Cannot invoke "ReadableInterval.getStartMillis()"   
    // because "validTime" is null  
}
```

Et le code dans tout ça ?

```
final ReadableInterval validTime = value.getValidTime();  
if (validTime.getStartMillis() < ...) {  
    // No exception here  
}
```

...

```
if (validTime.getStartMillis() < ...) {  
    // NullPointerException  
    // Cannot invoke "ReadableInterval.getStartMillis()"   
    // because "validTime" is null  
}
```

Un mauvais souvenir de Kafka

```
public void waitObject(Object obj, ...) throws InterruptedException {  
    synchronized (obj) {  
        while (true) {  
            if (condition.get())  
                return;  
            long currentTimeMs = milliseconds();  
            if (currentTimeMs >= deadlineMs)  
                throw new TimeoutException();  
            obj.wait(deadlineMs - currentTimeMs);  
            // wait() throws IllegalMonitorStateException  
        }  
    }  
}
```

Un mauvais souvenir de Kafka

```
public void waitObject(Object obj, ...) throws InterruptedException {  
    synchronized (obj) {  
        while (true) {  
            if (condition.get())  
                return;  
            long currentTimeMs = milliseconds();  
            if (currentTimeMs >= deadlineMs)  
                throw new TimeoutException();  
            obj.wait(deadlineMs - currentTimeMs);  
            // wait() throws IllegalMonitorStateException  
        }  
    }  
}
```

Un mauvais souvenir de Kafka

```
public void waitObject(Object obj, ...) throws InterruptedException {  
    synchronized (obj) {  
        while (true) {  
            if (condition.get())  
                return;  
            long currentTimeMs = milliseconds();  
            if (currentTimeMs >= deadlineMs)  
                throw new TimeoutException();  
            obj.wait(deadlineMs - currentTimeMs);  
            // wait() throws IllegalMonitorStateException  
        }  
    }  
}
```

KAFKA-10902 → JDK-8298446

Revenons à nos moutons

```
final ReadableInterval validTime = value.getValidTime();  
if (validTime.getStartMillis() < ...) {  
    // No exception here  
}
```

...

```
if (validTime.getStartMillis() < ...) {  
    // NullPointerException  
    // Cannot invoke "ReadableInterval.getStartMillis()"   
    // because "validTime" is null  
}
```

**Et je fais
quoi
maintenant
?!?**



Tu bliffes Martoni

- Ce n'est pas possible que ça arrive
- Debug « à l'ancienne »
- On voit bien une valeur et puis null
- Ce n'est pas du bluff

**Et je fais
quoi
maintenant
?!?**



Comment je sais si ça marche ?

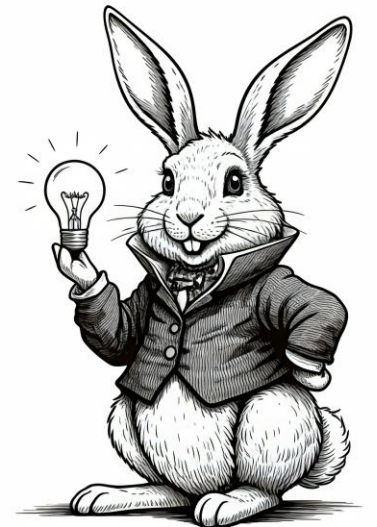
- Si ça plante c'est facile, mais si ça ne plante pas...
- Idée: 10 exécutions qui marchent alors c'est bon
 - probabilité de $1/1024$ de se tromper, $\sim 0,1\%$

**Et je fais
quoi
maintenant
?!?**



Idée n°1 – Problème de serveur

- Le benchmark tourne toujours sur le même serveur
- Si ça se trouve, il y a un problème avec le serveur
 - la mémoire,
 - le CPU,
 - le noyau linux,
 - la version de docker
 - ...
- Ce n'est pas un problème de serveur

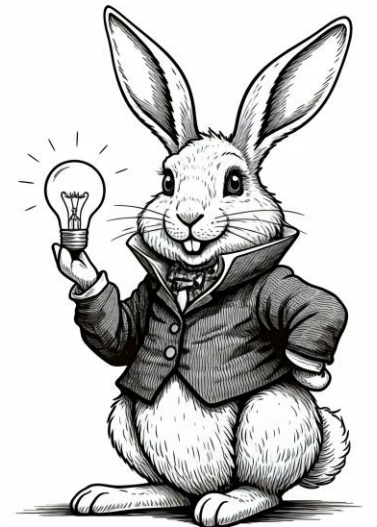


**Et je fais
quoi
maintenant
?!?**



Idée n°2 – Problème de simulateur

- On doit générer un jeu de données invalides
- Je rejoue les données d'un run qui passe, ça plante...
- ... et inversement !
- Ce n'est pas un problème de données



**Et je fais
quoi
maintenant
?!?**



Idée n°3 – Le benchmark n'est pas bon

- Le benchmark utilise un agent Java
- Si ça se trouve, l'agent n'aime pas Java 21
- Ce n'est donc pas un problème de benchmark



**Et je fais
quoi
maintenant
?!?**



Idée n°4 – Un bug de JVM ?

- Adoptium 17.0.8 ✓
- Adoptium 21.0.1 ✗
- Correto 21.0.1 ✗
- Oracle 21.0.1 ✗
- Adoptium 21 ✗
- Adoptium 17.0.9 ✓
- Adoptium 21 nightly ✗
- Latest (23) ✗
- Aleksey Shipilëv 21 builds
 - release ✗
 - fastdebug ✗
 - slowdebug ✗
- Adoptium 19.0.2 ✓

**Et je fais
quoi
maintenant
?!?**



Idée n°5 – Je ne suis pas le seul

- Chercher « NullPointerException » n'aide pas
- Même avec le numéro de version
- Même avec un filtre par date
- Rien sur la mailing list
- Rien sur le bug tracker

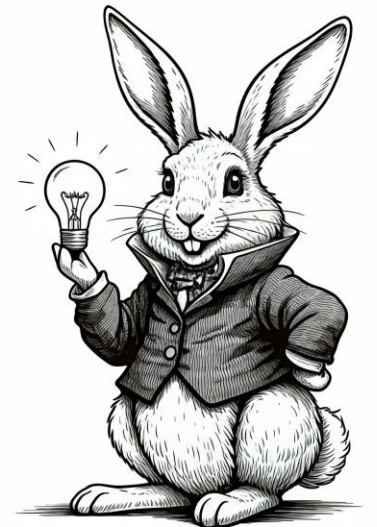


**Et je fais
quoi
maintenant
?!?**



Idée n°6 – Un reprocase

- Je ne peux pas envoyer mon code propriétaire
- J'essaie de créer un reprocase
- J'ai perdu une semaine...



**Et je fais
quoi
maintenant
?!?**



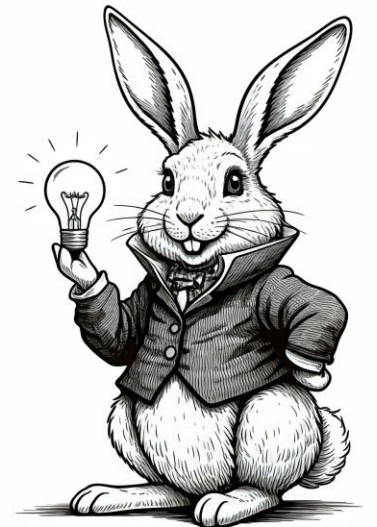
**Pas le choix,
je dois l'analyser moi-même**

**Et je fais
quoi
maintenant
?!?**



Idée n°7 – Le GC libère trop tôt ?

- Si ça se trouve, le GC libère l'objet trop tôt ?
- Tests avec les différents GC
 - -XX:+UseParallelGC X
 - -XX:+UseG1GC X
 - -XX:+UseZGC X
 - -XX:+UseShenandoahGC X
 - -XX:+UseSerialGC X
- Ce n'est donc pas le GC



**Et je fais
quoi
maintenant
?!?**

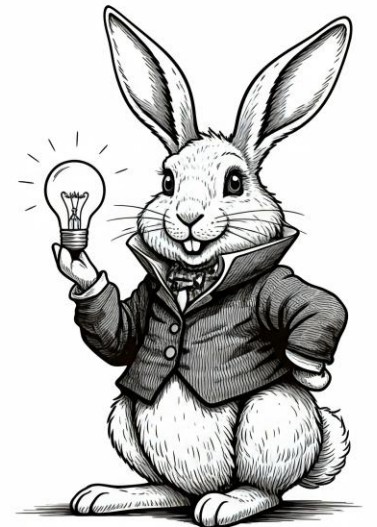


Idée n°8 – Parfois le JIT génère des bugs

- Comme ça plante vers la fin, ça peut venir du code compilé par le JIT
- Comme ça ne plante pas toujours, le code généré doit être faux 50% du temps
- Je n'ai qu'à comparer 2 exécutions

FAUX !

(mais je ne le sais pas encore)



Comment récupérer le code généré ?

- `-XX:+UnlockDiagnosticVMOptions`
- `-XX:CompileCommand="print package.Class::method"`
- `-XX:PrintAssemblyOptions=intel`
- **OpenJDK hsd**
 - disassembly plugin
 - dans le JDK ou en variable d'environnement
 - merci à Chris Newland pour sa compilation de la librairie

Ca ressemble à quoi ?

```
0x0000709395ed0480:  mov     DWORD PTR [rsp-0x14000],eax
0x0000709395ed0487:  push    rbp
0x0000709395ed0488:  sub     rsp,0x130
0x0000709395ed048f:  nop
0x0000709395ed0490:  cmp     DWORD PTR [r15+0x20],0x0
0x0000709395ed0498:  je      0x0000709395ed049f
0x0000709395ed049a:  call    Stub::nmethod_entry_barrier ; {runtime_call StubRoutines (final stubs)}
0x0000709395ed049f:  mov     QWORD PTR [rsp+0x80],rsi
0x0000709395ed04a7:  mov     QWORD PTR [rsp+0x70],rdx
0x0000709395ed04ac:  mov     QWORD PTR [rsp+0x88],rcx
0x0000709395ed04b4:  mov     DWORD PTR [rsp+0x68],r8d
0x0000709395ed04b9:  mov     DWORD PTR [rsp+0x98],r9d
0x0000709395ed04c1:  mov     DWORD PTR [rsp+0x6c],edi
0x0000709395ed04c5:  movabs  rbx,0x708f09311508
0x0000709395ed04cf:  mov     eax,DWORD PTR [rbx+0x8]
0x0000709395ed04d2:  add     eax,0x2
0x0000709395ed04d5:  mov     DWORD PTR [rbx+0x8],eax
0x0000709395ed04d8:  and     eax,0xffe
0x0000709395ed04dd:  test    eax,eax
0x0000709395ed04df:  je      0x0000709395ed1220
0x0000709395ed04e5:  mov     ebx,DWORD PTR [rdx+0x18] ; implicit exception: dispatches to 0x0000709395ed1241
```

...

Premières analyses

- Il y a beaucoup trop de lignes
- Les index à gauche ne correspondent pas
- Les pointeurs ne correspondent pas
- Impossible de comparer automatiquement

Les en-têtes de l'assembleur

```
===== C1-compiled nmethod =====  
----- Assembly -----
```

```
Compiled method (c1) 1107312 37562      2      package.Class::method (530 bytes)  
  total in heap  [0x0000709395ecff90,0x0000709395ed2480] = 9456
```

```
...
```

```
[Disassembly]
```

```
...
```

```
[/Disassembly]
```

Les en-têtes de l'assembleur

```
===== C1-compiled nmethod =====  
----- Assembly -----
```

```
Compiled method (c1) 1107312 37562      2      package.Class::method (530 bytes)  
  total in heap  [0x0000709395ecff90,0x0000709395ed2480] = 9456
```

...

```
[Disassembly]
```

...

```
[/Disassembly]
```

Ça passe !

Les en-têtes de l'assembleur

```
===== C2-compiled nmethod =====  
----- Assembly -----
```

```
Compiled method (c2) 1142813 38977      4      package.Class::method (530 bytes)  
  total in heap [0x000070939eece990,0x000070939eed88c8] = 40760
```

```
...
```

```
[Disassembly]
```

```
...
```

```
[/Disassembly]
```

Les en-têtes de l'assembleur

```
===== C2-compiled nmethod =====  
----- Assembly -----
```

```
Compiled method (c2) 1142813 38977      4      package.Class::method (530 bytes)  
  total in heap  [0x000070939eece990,0x000070939eed88c8] = 40760
```

...

```
[Disassembly]
```

...

```
[/Disassembly]
```

Ça plante !

Les en-têtes de l'assembleur

```
===== C2-compiled nmethod =====  
----- Assembly -----
```

```
Compiled method (c2) 1142813 38977          4      package.Class::method (530 bytes)  
  total in heap  [0x000070939eece990,0x000070939eed88c8] = 40760
```

...

```
[Disassembly]
```

...

```
[/Disassembly]
```

40ko !!!

(moi, je ne compare pas ça)

Les différentes options du JIT

- Aucun JIT
 - -Xint ✓
 - 35 minutes au lieu de 1 minute 30
- C1
 - -XX:TieredStopAtLevel=1 ✓
 - -XX:TieredStopAtLevel=2 ✓
 - -XX:TieredStopAtLevel=3 ✓
- C2 ✗

**Et je fais
quoi
maintenant
?!?**

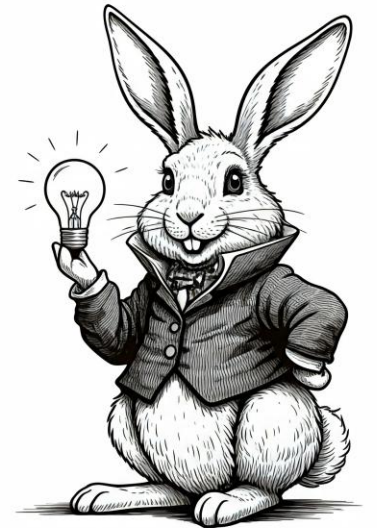


Idée n°9 – Test trop court ?

- Et si le plantage n'était pas aléatoire ?
- Si le benchmark n'avait pas le temps de planter ?
- Au lieu de rejouer 30 minutes de données, je tente 2 heures

Victoire !

(enfin... ça plante à tous les coups)

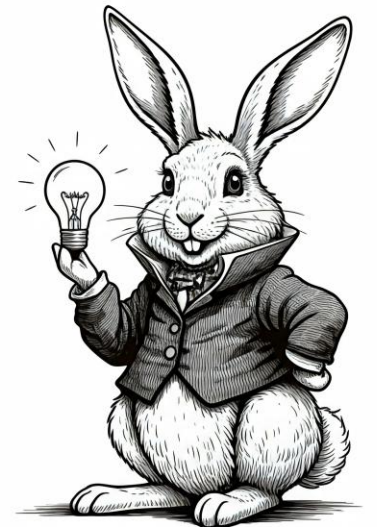


**Et je fais
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Idée n°10 – Et si je trouvais le commit ?

- Je vais utiliser git bisect pour trouver le commit responsable de la régression
- Ah, mais comment on build Java ?



Comment builder Java ?

```
$ git clone https://github.com/openjdk/jdk.git  
Cloning into 'jdk'...
```

```
$ cd jdk
```

```
$ git checkout jdk-21-ga  
Updating files: 100% (27123/27123), done.
```

Comment builder Java ?

```
$ bash configure
```

```
...
```

```
Autoconf is not found on the PATH
```

```
You might be able to fix this by running 'sudo apt-get  
install autoconf'.
```

```
$ sudo apt-get install autoconf
```

Comment builder Java ?

```
$ bash configure
```

```
Using autoconf at /usr/bin/autoconf
```

```
...
```

```
configure: error: Could not find required tool for FILE
```

```
$ sudo apt-get install file
```

Comment builder Java ?

```
$ bash configure
checking for file... /usr/bin/file
...
configure: error: Cannot find GNU make 3.81 or newer!
Please put it in the path, or add e.g.
MAKE=/opt/gmake3.81/make as argument to configure.
```

```
$ sudo apt-get install make
$ sudo apt-get install unzip
$ sudo apt-get install zip
```

Comment builder Java ?

```
$ bash configure
```

```
...
```

```
configure: Could not find a valid Boot JDK. OpenJDK  
distributions are available at http://jdk.java.net/.
```

```
configure: This might be fixed by explicitly setting  
--with-boot-jdk
```

Comment builder Java ?

```
$ bash configure --with-boot-jdk=/opt/jdk-20
```

```
...
```

```
configure: error: Could not find a C compiler. You might  
be able to fix this by running 'sudo apt-get install  
build-essential'.
```

```
$ sudo apt-get install build-essential
```

```
$ sudo apt-get install ...
```

Comment builder Java ?

```
$ bash configure --with-boot-jdk=/opt/jdk-20
```

...

A new configuration has been successfully created in
/home/user/jdk/build/linux-x86_64-server-release
using configure arguments '--with-boot-jdk=/opt/jdk-20'.

Configuration summary:

* Name: linux-x86_64-server-release

* Debug level: release

...

Comment builder Java ?

```
$ make images
```

```
...
```

```
Finished building target 'images' in configuration 'linux-x86_64-server-release'
```

```
$ cd build/linux-x86_64-server-release/jdk/bin
```

```
$ ./java --version
```

```
openjdk 21-internal 2023-09-19
```

Comment builder Java ?

- On installe tout ce dont il a besoin
 - outils, librairies X11, librairies son, ...
 - il fournit même les commandes à taper
- 5 minutes pour installer toutes les dépendances
- 5 minutes pour builder Java
- Facile de créer un environnement tout prêt

A la recherche du commit

```
[jdk-21-ga] $ git bisect start
```

```
[jdk-21-ga] $ git bisect bad
```

```
status: waiting for good commit(s), bad commit known
```

```
[jdk-21-ga] $ git bisect good jdk-19-ga
```

```
Bisecting: 2441 revisions left to test after this (roughly  
11 steps)
```

```
[2ccdefc81c0ea2ea5c4380bb045aa82ad1eb8205] 8299470: ...
```

A la recherche du commit

```
[2ccdefc81c0] $ make images
```

```
[2ccdefc81c0] $ ~/bench/test.sh
```

Crash!

```
[2ccdefc81c0] $ git bisect bad
```

```
Bisecting: 1219 revisions left to test after this (roughly  
10 steps)
```

```
[090cdfc7a2e280c620a0926512fb67f0ce7f3c21] 8294726: ...
```

A la recherche du commit

```
[090cdfc7a2e] $ make images
```

```
[090cdfc7a2e] $ ~/bench/test.sh
```

OK!

```
[090cdfc7a2e] $ git bisect good
```

```
Bisecting: 609 revisions left to test after this (roughly  
9 steps)
```

```
[761a4f4852cbb40660b6fb9eda4d740464218f75] 8295788: ...
```

A la recherche du commit

```
[761a4f4852c] $ make images
```

```
[761a4f4852c] $ ~/bench/test.sh
```

OK!

```
[761a4f4852c] $ git bisect good
```

```
Bisecting: 304 revisions left to test after this (roughly  
8 steps)
```

```
[27bbe7be2c43a22e8cf55aa403d8018346ae3e37] 8297976: ...
```

A la recherche du commit

```
[27bbe7be2c4] $ make images
```

```
[27bbe7be2c4] $ ~/bench/test.sh
```

OK!

```
[27bbe7be2c4] $ git bisect good
```

```
Bisecting: 152 revisions left to test after this (roughly  
7 steps)
```

```
[f4caaca100d334b671eed56287dfe7a1009c47d7] 8298852: ...
```

A la recherche du commit

```
[f4caaca100d] $ make images
```

```
[f4caaca100d] $ ~/bench/test.sh
```

OK!

```
[f4caaca100d] $ git bisect good
```

Bisecting: 73 revisions left to test after this (roughly 6 steps)

```
[34cdda5b8359cce33c2d4f92a41a620aea4f96e7] Merge
```

A la recherche du commit

```
[34cdda5b835] $ make images
```

```
[34cdda5b835] $ ~/bench/test.sh
```

Crash!

```
[34cdda5b835] $ git bisect bad
```

Bisecting: 42 revisions left to test after this (roughly 5 steps)

```
[e5edb10dc56d9edac8e050e0f8e6c116743975d6] 8299018: ...
```

A la recherche du commit

```
[e5edb10dc56] $ make images
```

```
[e5edb10dc56] $ ~/bench/test.sh
```

Crash!

```
[e5edb10dc56] $ git bisect bad
```

Bisecting: 17 revisions left to test after this (roughly 4 steps)

```
[5e678f7500e514f04637c546959613d4688f989c] 8298824: ...
```

A la recherche du commit

```
[5e678f7500e] $ make images
```

```
[5e678f7500e] $ ~/bench/test.sh
```

Crash!

```
[5e678f7500e] $ git bisect bad
```

Bisecting: 8 revisions left to test after this (roughly 3 steps)

```
[0ba473489151d74c8a15b75ff4964ac480fecb28] 8287699: ...
```

A la recherche du commit

```
[0ba47348915] $ make images
```

```
[0ba47348915] $ ~/bench/test.sh
```

OK!

```
[0ba47348915] $ git bisect good
```

Bisecting: 4 revisions left to test after this (roughly 2 steps)

```
[0eeae8b8e7ba40be5e93eb87c7e3dc94230062746] 8298808: ...
```

A la recherche du commit

```
[0eeae8b8e7ba] $ make images
```

```
[0eeae8b8e7ba] $ ~/bench/test.sh
```

Crash!

```
[0eeae8b8e7ba] $ git bisect bad
```

Bisecting: 1 revision left to test after this (roughly 1 step)

```
[3696711efa566fb776d6923da86e17b0e1e22964] Merge
```

A la recherche du commit

```
[3696711efa5] $ make images
```

```
[3696711efa5] $ ~/bench/test.sh
```

Crash!

```
[3696711efa5] $ git bisect bad
```

```
Bisecting: 0 revisions left to test after this (roughly 0  
steps)
```

```
[f771c56e16a39724712ca0d8c2dd55b9ce260f4d] 8298797: ...
```

A la recherche du commit

```
[f771c56e16a] $ make images
```

```
[f771c56e16a] $ ~/bench/test.sh
```

OK!

```
[f771c56e16a] $ git bisect good
```

```
3696711efa566fb776d6923da86e17b0e1e22964 is the first bad  
commit
```

```
Merge: f4caaca100d f771c56e16a
```

A la recherche du commit

```
[3696711efa5] $ git log --oneline --graph
```

```
*    3696711efa5 (HEAD) Merge
```

```
|\
```

```
| * f771c56e16a 8298797: ...
```

```
| * ...
```

```
| * 323e574a505 8298371: ...
```

```
* | f4caaca100d 8298852: ...
```

```
* | ...
```

```
* | 10bc86cc260 Merge
```

```
|\
```

A la recherche du commit

```
[3696711efa5] $ git bisect start
```

```
[3696711efa5] $ git bisect bad
```

```
status: waiting for good commit(s), bad commit known
```

```
[3696711efa5] $ git checkout 10bc86cc260
```

A la recherche du commit

```
[10bc86cc260] $ make images
```

```
[10bc86cc260] $ ~/bench/test.sh
```

OK!

```
[10bc86cc260] $ git bisect good
```

Bisecting: 15 revisions left to test after this (roughly 4 steps)

```
[e41686b4050d6b32fb451de8af39a78ec8bed0fd] 8298710: ...
```

A la recherche du commit

```
[e41686b4050] $ make images
```

```
[e41686b4050] $ ~/bench/test.sh
```

Crash!

```
[e41686b4050] $ git bisect bad
```

Bisecting: 7 revisions left to test after this (roughly 3 steps)

```
[0ef353925e645dd519e17aeb7a83e927271f8b95] 8298416: ...
```

A la recherche du commit

```
[0ef353925e6] $ make images
```

```
[0ef353925e6] $ ~/bench/test.sh
```

OK!

```
[0ef353925e6] $ git bisect good
```

Bisecting: 3 revisions left to test after this (roughly 2 steps)

```
[3cdbd878e68dc1131093137a7357710ad303ae8c] 8298241: ...
```

A la recherche du commit

```
[3cdbd878e68] $ make images
```

```
[3cdbd878e68] $ ~/bench/test.sh
```

Crash!

```
[3cdbd878e68] $ git bisect bad
```

Bisecting: 1 revision left to test after this (roughly 1 step)

```
[4b313b51b1787113961c289a41708e31fa19cacc] 8297798: ...
```

A la recherche du commit

```
[4b313b51b17] $ make images
```

```
[4b313b51b17] $ ~/bench/test.sh
```

OK!

```
[4b313b51b17] $ git bisect good
```

```
Bisecting: 0 revisions left to test after this (roughly 0  
steps)
```

```
[10737e168c967a08e257927251861bf2c14795ab] 8298468: ...
```

A la recherche du commit

```
[10737e168c9] $ make images
```

```
[10737e168c9] $ ~/bench/test.sh
```

Crash!

```
[10737e168c9] $ git bisect bad
```

```
10737e168c967a08e257927251861bf2c14795ab is the first bad  
commit
```

A la recherche du commit

commit 10737e168c967a08e257927251861bf2c14795ab

Author: xxx <xxx@openjdk.org>

Date: Thu Dec 15 19:54:25 2022 +0000

8298468: Clean up class_loader parameters

Reviewed-by: yyy, zzz

A la recherche du commit

<code>src/hotspot/share/cds/cdsProtectionDomain.cpp</code>		3
<code>src/hotspot/share/classfile/loaderConstraints.cpp</code>		70
<code>src/hotspot/share/classfile/loaderConstraints.hpp</code>		13
<code>src/hotspot/share/classfile/systemDictionary.cpp</code>		74
<code>src/hotspot/share/classfile/systemDictionary.hpp</code>		8

5 files changed, 76 insertions(+), 92 deletions(-)

Commit trouvé !

src/hotspot/share/classfile/loaderConstraints.cpp

+31 -39

```
@@ -47,11 +47,11 @@ class LoaderConstraint : public CHeapObj<mtClass> {
    // not class loaders.
    GrowableArray<ClassLoaderData*>* _loaders;           // initiating loaders
    public:
-   LoaderConstraint(InstanceKlass* klass, oop class_loader1, oop class_loader2) :
+   LoaderConstraint(InstanceKlass* klass, ClassLoaderData* loader1, ClassLoaderData* loader2) :
    _klass(klass) {
        _loaders = new (mtClass) GrowableArray<ClassLoaderData*>(10, mtClass);
-   add_loader(class_loader1);
-   add_loader(class_loader2);
+   add_loader_data(loader1);
+   add_loader_data(loader2);
    }
    LoaderConstraint(const LoaderConstraint& src) = delete;
    LoaderConstraint& operator=(const LoaderConstraint&) = delete;
@@ -61,7 +61,7 @@ class LoaderConstraint : public CHeapObj<mtClass> {
    InstanceKlass* klass() const { return _klass; }
    void set_klass(InstanceKlass* k) { _klass = k; }
```

Commit trouvé !

```
src/hotspot/share/classfile/loaderConstraints.cpp  +31 -39  ...
164 157 // Either add it to an existing entry in the table or make a new one.
165 - void LoaderConstraintTable::add_loader_constraint(Symbol* name, InstanceKlass* klass, oop class_loader1, oop class_loader2) {
158 + void LoaderConstraintTable::add_loader_constraint(Symbol* name, InstanceKlass* klass,
159 + ClassLoaderData* loader1, ClassLoaderData* loader2) {
166 160     assert_lock_strong(SystemDictionary_lock);
167 - LoaderConstraint* constraint = new LoaderConstraint(klass, class_loader1, class_loader2);
161 + LoaderConstraint* constraint = new LoaderConstraint(klass, loader1, loader2);
168 162
169 163     // The klass may be null if it hasn't been loaded yet, for instance while checking
170 164     // a parameter name to a method call. We impose this constraint that the
@@ -256,22 +250,22 @@ void LoaderConstraintTable::purge_loader_constraints() {
256 250 }
257 251
258 252 void log_ldr_constraint_msg(Symbol* class_name, const char* reason,
259 - Handle class_loader1, Handle class_loader2) {
253 + ClassLoaderData* loader1, ClassLoaderData* loader2) {
260 254     LogTarget(Info, class, loader, constraints) lt;
261 255     if (lt.is_enabled()) {
```

Commit trouvé !

src/hotspot/share/classfile/loaderConstraints.cpp

+31 -39

```
265 259      class_name->as_C_string(),
266 -      ClassLoaderData::class_loader_data(class_loader1())->loader_name_and_id(),
267 -      ClassLoaderData::class_loader_data(class_loader2())->loader_name_and_id(),
260 +      loader1->loader_name_and_id(),
261 +      loader2->loader_name_and_id(),
268 262      reason);
269 263  }
270 264  }
271 265
272 266  bool LoaderConstraintTable::add_entry(Symbol* class_name,
273 -      InstanceKlass* klass1, Handle class_loader1,
274 -      InstanceKlass* klass2, Handle class_loader2) {
267 +      InstanceKlass* klass1, ClassLoaderData* loader1,
268 +      InstanceKlass* klass2, ClassLoaderData* loader2) {
275 269
276 270      LogTarget(Info, class, loader, constraints) lt;
277 271      if (klass1 != NULL && klass2 != NULL) {
```

@@ -282,35 +276,35 @@ bool LoaderConstraintTable::add_entry(Symbol* class_name,

Le ticket associé



JDK / JDK-8298468

Clean up class_loader parameters

Resolved ▾

 Export ▾

▾ Details

Type:  Enhancement
Priority:  P4
Affects Version/s: 21
Component/s: hotspot
Labels: [classloading](#)

Resolution: Fixed
Fix Version/s: [21](#)

▾ Description

The LoaderConstraintTable stores ClassLoaderData, but passes class_loader objects. There's a lot of unnecessary conversion back and forth that can be cleaned up.

▾ Issue Links

links to

**Et je fais
quoi
maintenant
?!?**



Appeler à l'aide

- Commit author
- Aleksey Shipilëv
- Nicolai Parlog
- Rémi Forax
- Mailing list java



**Et je fais
quoi
maintenant
?!?**



Relecture du commit

```
src/hotspot/share/classfile/loaderConstraints.cpp  +31 -39 ...
```

```
348 340      } else if (pp1 == NULL) {
349 -    pp2->extend_loader_constraint(class_name, class_loader1, klass);
350 +    pp2->extend_loader_constraint(class_name, loader1, klass);
351 342      } else if (pp2 == NULL) {
352 -    pp1->extend_loader_constraint(class_name, class_loader2, klass);
353 +    pp1->extend_loader_constraint(class_name, loader1, klass);
354 344      } else {
355 345          merge_loader_constraints(class_name, pp1, pp2, klass);
356 346      }
@@ -359,7 +351,7 @@ bool LoaderConstraintTable::add_entry(Symbol* class_name,
// return true if the constraint was updated, false if the constraint is
// violated
bool LoaderConstraintTable::check_or_update(InstanceKlass* k,
-    Handle loader,
+    ClassLoaderData* loader,
    Symbol* name) {
    LogTarget(Info, class, loader, constraints) lt;
    LoaderConstraint* p = find_loader_constraint(name, loader);
```

Relecture du commit

- Ça ressemble à une typo
- Si je mets loader2, est-ce que ça marche ?

Victoire !

(reste plus qu'à l'intégrer)

Création de la pull request

- Signer le Oracle Contributor Agreement
 - partage de la propriété intellectuelle avec Oracle
- Il faut créer un bug sur le tracker d'abord
- J'ai demandé à Aleksey Shipilëv de le faire
 - JDK-8322282

Création du bug



JDK / JDK-8322282

Incorrect LoaderConstraintTable::add_entry after JDK-8298468

Closed ▾

▼ Details

Type:	 Bug	Resolution:	Fixed
Priority:	 P2	Fix Version/s:	23
Affects Version/s:	21, 22, 23		
Component/s:	hotspot		

▼ Description

As reported here by Antoine Dessaigne:

<https://mail.openjdk.org/pipermail/hotspot-compiler-dev/2023-December/071086.html>

There is a regression introduced by [JDK-8298468](#):

<https://github.com/openjdk/jdk/commit/10737e168c967a08e257927251861bf2c14795ab#diff-49d18b7f77db80ff67872c7db1e4c87e98a126e63fff5f63353e4406db332b19R343>

...which looks like a typo.

Création de la pull request

▼ ↕ 4 ■■■■■ src/hotspot/share/classfile/loaderConstraints.cpp 📄

☐ Viewed 🗨️ ⋮

...	...	@@ -1,5 +1,5 @@
1	1	/*
2		- * Copyright (c) 2003, 2023, Oracle and/or its affiliates. All rights reserved.
	2	+ * Copyright (c) 2003, 2024, Oracle and/or its affiliates. All rights reserved.
3	3	* DO NOT ALTER OR REMOVE COPYRIGHT NOTICES OR THIS FILE HEADER.
4	4	*
5	5	* This code is free software; you can redistribute it and/or modify it
⋮ ↓ ↑ ⋮		@@ -387,7 +387,7 @@ bool LoaderConstraintTable::add_entry(Symbol* class_name,
387	387	} else if (pp1 == nullptr) {
388	388	pp2->extend_loader_constraint(class_name, loader1, klass);
389	389	} else if (pp2 == nullptr) {
390		- pp1->extend_loader_constraint(class_name, loader1, klass);
	390	+ pp1->extend_loader_constraint(class_name, loader2, klass);
391	391	} else {
392	392	merge_loader_constraints(class_name, pp1, pp2, klass);
393	393	}
⋮ ↓		

**Corrigé
en 21.0.3**



Bilan

- J'étais perdu mais en fait c'était simple
- C'est très facile de compiler Java
 - Je m'en suis servi une seconde fois
- J'ai appris plein de choses au passage
 - les différents niveaux de JIT
 - comment voir l'assembleur généré
- Attention aux commits de refactor
 - même s'il y a une code review...

**Merci pour
votre écoute**



Antoine DESSAIGNE



Slides

